

the website they want to block to the list. When they try to visit that site, a message that states that the website is blocked will be displayed, and access to it will be denied. If the users then want to visit the site, they will have to unblock it by deleting the website entry from the list of blocked sites in the extension.

We are going to use the following Web technologies to build this extension:

- HTML to create a table to view blocked sites, and buttons to add and delete them
- CSS to style HTML
- JavaScript (Chrome browser API) to write our logic

Setting up the development environment

Let us now set up the file structure for our browser extension. Create a new folder and give it a name, say 'web-blocker'. Now, inside that folder, create the following empty files:

- *manifest.json*
- *options.html*
- *script.js*
- *block.js*
- *style.css*

You should end up with the directory structure shown in Figure 1.

- *manifest.json* is a metadata file in JSON format. It contains basic information about the extension, like its name, a description of it, the version number, which script it should run on starting up, etc.
- *options.html* displays the UI of the extension, like the list of websites to be blocked, buttons to add and delete, websites from the list, etc.
- *script.js* controls the interactions and the logic of the *options.html* file.

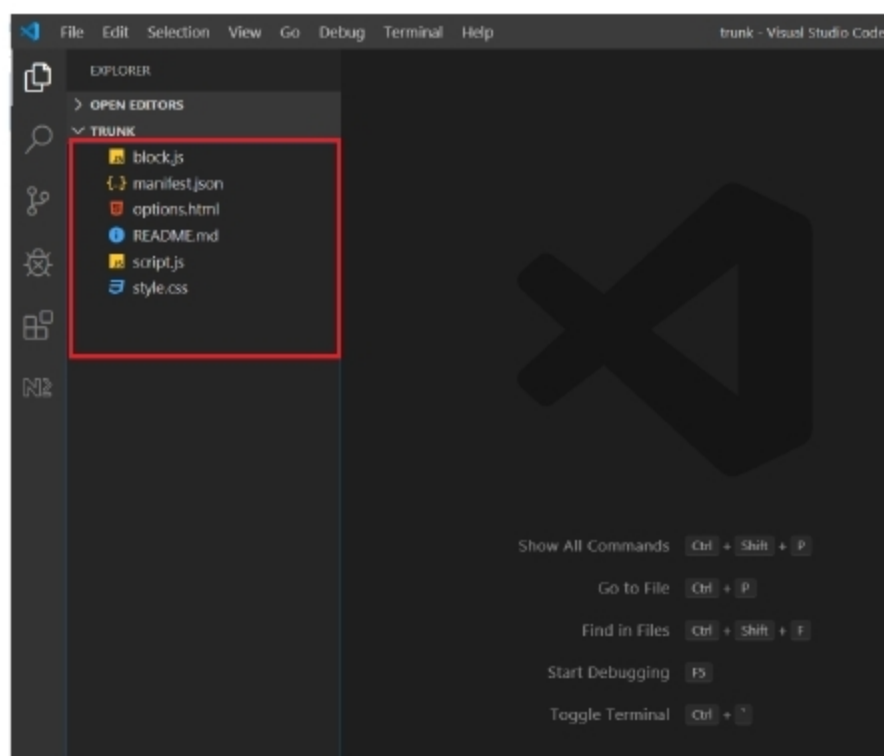


Figure 1: Extension project file structure

- *block.js*: In this file we are going to intercept the outgoing request from the browser. If the request contains one of the blocked websites from the list, we will reject the request, thus blocking that website.
- *style.css* adds styles to the *options.html* page to make it look good.

Creating the manifest file

Open the *manifest.json* file and copy the following JSON code:

```
{
  "name": "Website Blocker",
  "version": "1.0.0",
  "description": "Blocks websites based on domain names",
  "background": { "scripts": ["block.js"] },
  "options_page": "options.html",
  "permissions": [
    "webRequest", "webRequestBlocking",
    "http://**/*", "https://**/*",
    "storage"
  ],
  "manifest_version": 2
}
```

As you can see, this code is in JSON format. It consists of general information about the browser extension, such as its version, name, description, permissions required for the extension, and so on.

The parameter description is as follows:

- *manifest_version* specifies the version of the manifest; the value must be 2
- *name* specifies the name of the extension
- *version* specifies the version of the extension
- *description* specifies a short description of the extension
- *background* specifies that the script *block.js* should load as soon as the user starts the browser
- *options_page* specifies the UI page to be displayed for extension options
- *permission* specifies which browser APIs are to be used. We are using *webRequest*, *webRequestBlocking*, *https://**/**, *http://**/** and *storage*

Note: It is assumed that you have some basic knowledge of Web technologies like HTML, CSS and JavaScript. If you don't, W3Schools (<http://www.w3schools.com/>) is a good place to start. The site has some great tutorials for Web technologies that are easy to follow.

To add HTML, open the *options.html* file located in the extension directory, and add the following HTML code: